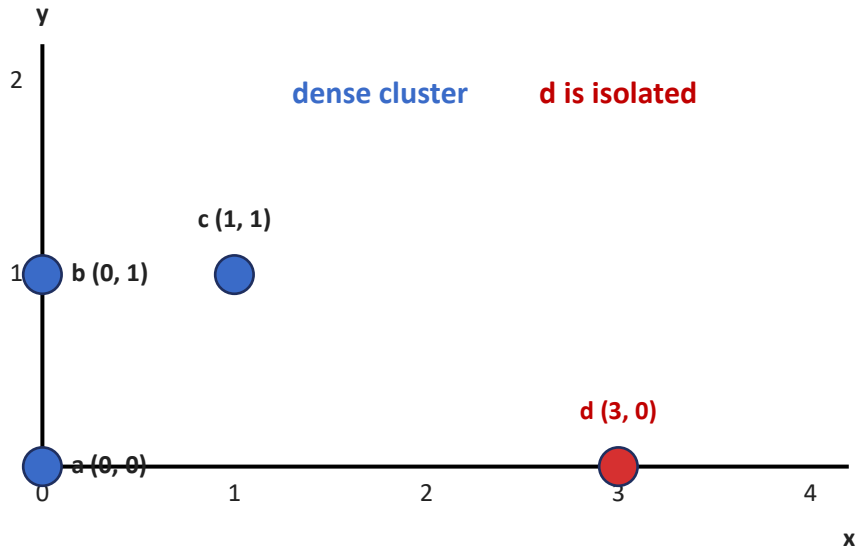
LOF Example (2D Data)

LOF Example: k=2 and Manhattan distance

Point	x	y
a	0	0
b	0	1
c	1	1
d	3	0



- Is d an outlier compared with a, b, c?
- We look at the 2 nearest neighbours of every point.

Step 1a: Distance Table

$$dist(p, q) = |x_1 - x_2| + |y_1 - y_2|$$

Point	x	y
a	0	0
b	0	1
c	1	1
d	3	0

	a	b	c	d
a	0	1	2	3
b	1	0	1	4
c	2	1	0	3
d	3	4	3	0

- Example: dist(a, b) = $|0 - 0| + |0 - 1| = 1$, dist(b, d) = $|0 - 3| + |1 - 0| = 4$
- dist(a, d) = 3 and dist(c, d) = $|1 - 3| + |1 - 0| = 3$

Step 1b: k-distance and Neighbourhood

Sort distances, pick the k=2 nearest; k-distance = the 2nd one

Point	Distances	Sorted distances	$N_2(p)$	2-distance(p)
a	b(1), c(2), d(3)	b(1), c(2), d(3)	b(1), c(2)	**2**
b	a(1), c(1), d(4)	a(1), c(1), d(4)	a(1), c(1)	**1**
c	a(2), b(1), d(3)	b(1), a(2), d(3)	b(1), a(2)	**2**
d	**a(3), b(4), c(3)**	**a(3), c(3), b(4)**	**a(3), c(3)**	**3**

- Example: d is closest to a(3) and c(3), so 2-distance(d) = dist(d, a) = 3.
- b has a tie: a(1) and c(1), so 2-distance(b) = 1 and $N_2(b) = \{a, c\}$.

Step 2: Reachability Distance

reach-dist(p, o) = max(k-distance(o), dist(p, o))

Point	$N_2(p)$	2-distance(p)
a	b(1), c(2)	2
b	a(1), c(1)	1
c	b(1), a(2)	2
d	a(3), c(3)	3

p → o	k-distance(o)	dist(p, o)	max = reach-dist
a → b	1	1	max(1, 1) = 1
a → c	2	2	max(2, 2) = 2
b → a	2	1	max(2, 1) = 2
b → c	2	1	max(2, 1) = 2
c → b	1	1	max(1, 1) = 1
c → a	2	2	max(2, 2) = 2
d → a	2	3	max(2, 3) = 3
d → c	2	3	max(2, 3) = 3

Step 3: Local Reachability Density

$lrd(p) = |N_2(p)| / \Sigma$ reach-dist with $|N_2(p)| = 2$

p → o	k-distance(o)	dist(p, o)	max = reach-dist
a → b	1	1	max(1, 1) = 1
a → c	2	2	max(2, 2) = 2
b → a	2	1	max(2, 1) = 2
b → c	2	1	max(2, 1) = 2
c → b	1	1	max(1, 1) = 1
c → a	2	2	max(2, 2) = 2
d → a	**2**	**3**	**max(2, 3) = 3**
d → c	**2**	**3**	**max(2, 3) = 3**

Point	Σ reach-dist	lrd = 2 / Σ	lrd value
a	1 + 2 = 3	2 / 3	0.667
b	2 + 2 = 4	2 / 4	0.5
c	1 + 2 = 3	2 / 3	0.667
d	**3 + 3 = 6**	**2 / 6**	**0.333**

- d has a much lower density (0.333) than a, b, c (0.5 to 0.667).

Step 4: LOF Score for All Points

LOF(p) = Avg lrd of $N_2(p)$ / lrd(p)

Point	lrd value
a	**0.667**
b	**0.5**
c	**0.667**
d	**0.333**

Point	$N_2(p)$	lrd(p)	Avg lrd of $N_2(p)$
a	{b, c}	0.667	[lrd(b)+lrd(c)]/2 → (0.5 + 0.667) / 2 = 0.583
b	{a, c}	0.5	[lrd(a)+lrd(c)]/2 → (0.667 + 0.667) / 2 = 0.667
c	{b, a}	0.667	[lrd(b)+lrd(a)]/2 → (0.5 + 0.667) / 2 = 0.583
d	**{a, c}**	**0.333**	**[lrd(a)+lrd(c)]/2 → (0.667 + 0.667) / 2 = 0.667**

- Neighbour sets: $N_2(a) = \{b, c\}$, $N_2(b) = \{a, c\}$, $N_2(c) = \{b, a\}$, $N_2(d) = \{a, c\}$.

Step 4: LOF Score for All Points

LOF(p) = Avg lrd of $N_2(p)$ / lrd(p)

Point	$N_2(p)$	lrd(p)	Avg lrd of $N_2(p)$	LOF
a	{b, c}	0.667	[lrd(b)+lrd(c)]/2 → (0.5 + 0.667) / 2 = 0.583	Avg lrd / lrd(a) → 0.583 / 0.667 ≈ 0.88
b	{a, c}	0.5	[lrd(a)+lrd(c)]/2 → (0.667 + 0.667) / 2 = 0.667	Avg lrd / lrd(b) → 0.667 / 0.5 ≈ 1.33
c	{b, a}	0.667	[lrd(b)+lrd(a)]/2 → (0.5 + 0.667) / 2 = 0.583	Avg lrd / lrd(c) → 0.583 / 0.667 ≈ 0.88
d	**{a, c}**	**0.333**	**[lrd(a)+lrd(c)]/2 → (0.667 + 0.667) / 2 = 0.667**	**Avg lrd / lrd(d) → 0.667 / 0.333 ≈ 2.00**

- Neighbour sets: $N_2(a) = \{b, c\}$, $N_2(b) = \{a, c\}$, $N_2(c) = \{b, a\}$, $N_2(d) = \{a, c\}$.
- LOF is computed from unrounded lrd values, so the last digit can differ slightly.

Final Score Comparison

LOF(p) = average lrd of neighbours / lrd(p)

Point	lrd(p)	Avg lrd of $N_2(p)$	LOF	Result
a	0.667	0.583	**0.88**	Normal
b	0.5	0.667	**1.33**	Normal
c	0.667	0.583	**0.88**	Normal
d	**0.333**	**0.667**	**2.00**	**Outlier**

- d: LOF ≈ 2.00, so it is about 2 times sparser than its neighbours: the top-1 outlier.
- a, b, c: LOF close to 1 (0.88 to 1.33), so they are normal. Ranking: d > b > a = c.
- Note: the product form Σlrd(o') × Σreach-dist gives 4× these values (d = 8.004, b = 5.336, a = c = 3.501): same ranking.